

Interview Questions

1) What is LLM?

- LLM stands for Large Language Model.
- LLM are deep neural network trained on massive text data to understand & generate human-like text.
- Real time example :- chatbot / virtual assistant.

A user types "What is capital of India?"

LLM processes the query via an API (OpenAI, Hugging face) & returns "The capital of India is Delhi."

→ Uses in Real projects :-

- 1) chatbots & copilots
- 2) RAG systems

2) What is a token?

- A token is the smallest unit of text that a model can understand. It can be a word, subword, or even a single character.
- Example :- "Unbelievable" → "un", "believ", "able".

3) What is tokenization?

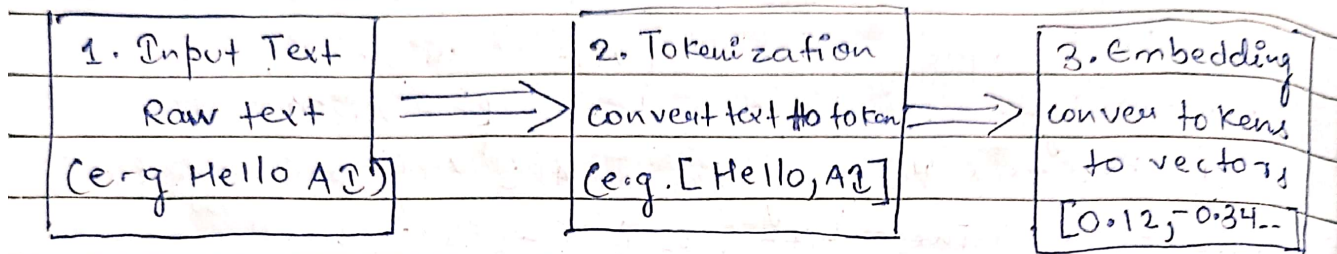
- It is the process of converting text into tokens so that the model can process it.
- Common tokenizers : BPE, Wordpiece, Sentencepiece.

4) What is an embedding?

- It is a numerical representation of text (or other data) in a continuous vector space.
- It helps the model understand semantic meaning.

Example:- The words "king" & "Queen" will have vectors that are close to each other.

Tokenization to Embedding (3-step flow)



5) What is Text Normalization?

- It is the process of converting text into standard or canonical format by removing noise & variations.
- It cleans & standardizes raw text so that downstream GenAI models can process it consistently.
- It is the first & most critical step in any pipeline.

→ Real time examples :-

- chatbot preprocessing

A customer support chatbot receive:

"I'm not able 2 access my account!!!! @_@"

After Normalization:

"I am not able to access my account."

→ Uses in Real projects:-

- 1) chatbots / conversational AI
- 2) Search Engines.

6) what is NLP?

→ Natural language processing is a subfield of AI that enables computers to understand, interpret, and generate human language.

→ The fundamental building blocks include tokenization, stopword removal, stemming & lemmatization.

→ Real time Examples :- Search Engine Indexing

Document : "Organization activates connection b/w studies"

stemming reduces all to root forms : ["organ", "activ", "connect", "between", "studi"]

This allows matching queries like "Organize", "Organizing", "Organization" to the same document.

→ Uses in Real projects :-

1) Search Engines

2) Spam detection.

7) what is NLTK?

→ It stands for Natural Language toolkit

→ It is a python library which uses tokenization by default.

→ Real time examples :-

1) Chatbots

2) Spam detection

8) Diffⁿ b/w Stemming & Lemmatization?

Stemming

→ It cuts a word down to its root form

lemmatization

→ It finds the meaningful dictionary form.

→ It Uses Rule-based Method.	→ It uses vocabulary / linguistic analysis.
→ It is less accurate	→ It is more accurate
→ It is faster	→ It is slower
→ Output of stemming may not be a real word.	→ Output of lemmatization Usually a real word.
→ E.g:- Studies → Studi?	→ E.g:- Studies → Study.

9) What is POS tagging & why ~~is~~ is it important for lemmatization?

- A POS (Part of Speech) tagging identifies if a word is a noun, verb, adjective etc.
- Lemmatization requires POS context because "running" as a noun (the running) vs verb (running fast) lemmatizes differently.

lemmatize("running", "n") vs lemmatize("running", "v")

10) What are stop words & how do you handle them?

- Stop words are common words (the, a, is, in, and) that carry little semantic meaning.
- They are removed to reduce noise & focus on meaningful words. It handles depends on specific application & preprocessing pipeline.

10) What is chunking & why it is needed in GenAI?

→ It is the process of splitting large document into smaller, ~~manageable~~ manageable pieces while preserving semantic meaning.

→ It is needed in GenAI because it is critical in RAG because 1) embedding models have max token limits, 2) LLM context windows are finite, 3) retrieval is more precise with smaller chunks, so in last 4) It reduces the cost by sending only relevant context.

→ Real time example -

- 1) RAG System for Customer Support
- 2) Processing Legal documents

→ Uses in projects -

- 1) RAG pipelines
- 2) LLM context windows

→ Types of chunking -

- 1) Fixed Size (split by character count with overlap)
- 2) Recursive (split hierarchically by separators)
- 3) ~~Split by topic/semantic~~
- 3) Semantic (split by topic / semantic boundaries)
- 4) Document-Based (split by document structure).

11) Diffⁿ b/w Fixed Size & Recursive chunking

Fixed Size	Recursive
→ It splits text into chunks based mainly on fixed no. of char.	→ It tries to split text using Hierarchical separators.
→ It may ignore structure.	→ It tries to preserve structure
→ It is very simple implementation.	→ It is more sophisticated implementation.
→ It is faster	→ It is slower
→ It is Good for RAG Suitability.	→ It is often better for documents
e.g. Every 500 tokens	e.g. Paragraph → sentence → word.

12) what is RAG (Retrieval-Augmented Generation)?

→ It is a technique in GenAI that improves the quality & accuracy of responses by allowing LLM (Large Language Model) to retrieve relevant information before generating an answer.

→ How RAG Works:-

User query

↓
Query Embedding

↓
Vector Database

↓
Retrieved Documents

↓
LLM + user query + Retrieved context

↓
Generated Response.

13) What is prompt Engineering?

→ It is the practice of designing & optimizing text inputs to AI models to get the best possible outputs.

→ It is the interface b/w human intent & machine intelligence.

→ Where it is used :-
chat bots & virtual assistant.

→ Real world application :-

Git hub, Copilot uses prompts to generate code.

→ There are 8 types of Prompt Engineering :-

1) Zero-shot prompting

→ You ask the AI to perform without providing any prior examples.

→ It is the simplest & fastest approach.

→ Best Use cases :-

Simple classification, summarization, translation, question answering.

2) One-shot prompting

→ You provide exactly one example before asking the actual question.

→ When it doesn't give a consistent output format, one example often does the trick.

→ Best Use cases :-

Email or message templates with a fixed structure.

3) Few-shot prompting

→ You provide 2-5 examples of input - output pairs before asking the actual question.

→ Best use cases:-

1) Sentiment analysis & text classification

4) Chain-of-thought prompting

→ You ask the model to solve a problem through step-by-step reasoning before giving a final answer.

→ Best use cases:-

1) Math word problems & quantitative reasoning.

5) Role Based prompting

→ You assign a specific role/persona, job title to the AI before asking your question

→ Best use cases:-

1) Interview prep with a realistic interviewer persona.

6) Instruction Based prompting

→ You give clear instructions about what the model should do.

→ When you need highly specific, structured outputs.

→ Best use cases:-

1) Generating structured data (JSON, csv, tables)

7) Contextual prompting

→ You provide relevant background information/context before asking the question.

→ Best use cases :-

1) chatbots

2) business assistants

8) Step-by-step prompting

→ It means giving an AI model a complex task in a clear sequence of instructions so it can handle each part systematically.

→ Best use cases :-

1) Project planning & workflow breakdowns.

14) What is temperature in LLM?

→ Temperature controls randomness, when we have low (0-0.3) deterministic, factual &

when we have high (0.7-1.0) creative, diverse.

For RAG use low temperature, for creative writing use higher temperature.

→ At temperature = 0, the model always picks the most likely token.

15) Diffⁿ b/w Embedding & Generative Model

→ An Embedding model converts text to fixed size vectors (for search similarity) whereas Generative model produces new text tokens.

→ In RAG, you use both an embedding model to index/retrieve & generative model to answer.

16) What are Vector Embeddings & Vector Databases?

→ Embeddings → Numerical vectors representing text / images.

→ Similar items have similar vectors (Cosine Similarity)

→ Vector DB → Stores embedding & enables fast similarity search.

→ Essential for RAG.

17) Explain the RAG architecture Step by step

1) Index :- Load documents → chunk → Embed → store in vector DB

2) Retrieve :- User query → Embed query → Search Vector DB for similar chunks.

3) Generate :- Build prompt with "Context [chunks] + Question: [query]" → LLM generates answer.

Q) 18) What is LCEL?

→ LCEL stands for Langchain expression language

→ It combines multiple steps through chains with the help of ("|") pipeline symbol.

→ For example :-

Step-1 → Structured prompt

Step-2 → LLM

Step-3 → Output parser

Prompt



LLM



Output parser

→ Without LCEL, connecting multiple AI components can require more code that's why we use LCEL. This creates a complete AI pipeline.

→ Main components of LCEL :-

1) Prompt template

→ Create instructions for the AI model

2) LLM (Large Language Model)

→ The AI model processes the prompt.

E.g :- Google Gemini, Open AI GPT.

3) Output parser

→ Converts the model response into the required format.

→ The ("|") pipeline symbol means :-

Take the output from one component and send it to the next component.

→ Real time Example :- AI student Assistant

•) Imagine you are building an AI application for students

The student asks :- "Explain ML"

•) The LCEL workflow will be :-

Student Question



prompt template



Gemini GPT model



Output parser



Simple answer

→ Best Use Cases of LCEL :-

- 1) chatbots
- 2) RAG application
- 3) Resume Analyzer
- 4) AI Agents

19) What is Retrieval testing?

→ It is a part of RAG architecture & it checks the chunks which is retrieved from vector database by using similarity search.

→ Here we use 4th methods:-

- 1) Precision @ K
- 2) Recall @ K
- 3) Mean Reciprocal Rank (MRR)
- 4) Hit rate

1) Precision @ K

→ Out of all the retrieve chunks how many % of checks chunks are correctly retrieve

→ Formula:-
$$\frac{\text{correctly retrieve chunks}}{K (\text{Total Retrieve chunks})}$$

2) Recall @ K

→ It checks out of all the actual chunks how many % of chunks are correctly retrieve

→ Formula =
$$\frac{\text{Correctly retrieve chunks}}{K (\text{Total Retrieve chunks})}$$

3) MRR (Mean Reciprocal Rank) :-

→ It checks Retrieval testing score over all performance. ~~for all the~~

4) Hit rate

→ It checks if atleast one chunk from retrieved chunks is present in the relevant chunks it returns ~~only 0 or 1~~ 1.

→ If no chunk present in the retrieved chunk then it gives 0.

→ Best use cases :- Company HR chatbot.

•) Imagine a company creates an HR chatbot.

•) The company has document about :-

Leave policy

WFH ~~policy~~

Salary policy

Holidays

•) Employee asks :-

"How many WFH days are allowed?"

•) The ~~the~~ RAG system should retrieve :-

WFH policy

•) Retrieval testing checks :-

Question

↓
retriever

Did it retrieve WFH policy? ✓

•) If it retrieves :-

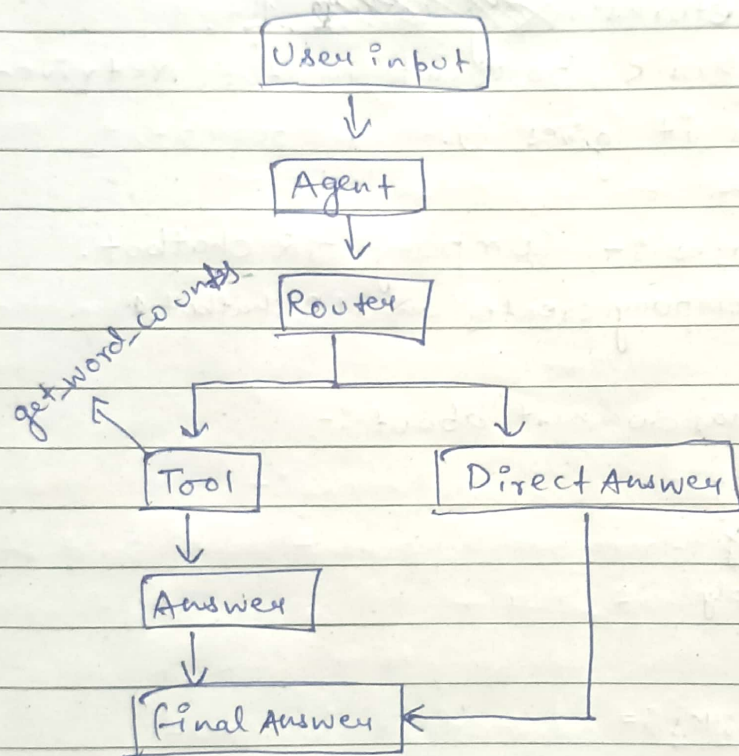
Salary policy X

Then the retrieval system needs improvement.

20) What is Agentic AI ?

→ It is a GenAI system that doesn't generate ~~answer~~ answer, it can decide what actions are needed to achieve a goal and execute those actions.

→ Work flow :-



→ Components of Agentic AI :-

1) LLM :-

→ It acts as the decision-making engine.
e.g. GPT, Gemini, Claude

→ It understands the user's goal & helps decide what to do next.

2) Tools :-

→ Agents can use external ~~tools~~ tools.

e.g.:- Search tool, Python.

3) Memory :-

→ It allows an agent to retain useful information.

4) Planning :-

→ The agent breaks a large goal into smaller tasks.

5) Action :-

→ The agent actually performs tasks using tools.

6) Observation :-

→ After using tool, the agent examines the result & decides what to do next.

→ Real time example :- AI Travel Agent

Step-1 Understand the goal

Step-2 Create a plan

Step-3 Use tools

Step-4 Evaluate options

Step-5 Make decisions

Step-6 Take action

Step-7 Report

→ Use cases :-

1) Customer support

2) Software development

3) Data analysis.